
KYLE DICK

GUELPH, ON | 519-760-3624 | KDICK6287@CONESTOGAC.ON.CA | ELECTRONIC
SYSTEMS ENGINEERING B.ENG. CANDIDATE | EXPECTED GRADUATION: AUG. 2026

OBJECTIVE

Electronic Systems Engineering B.Eng. candidate with hands-on experience developing embedded firmware for ESP32-S3 and STM32 systems, integrating sensors and communication interfaces, and connecting edge devices to Raspberry Pi/Linux services. Strongest in C, with working knowledge of C++ and Python, and experience debugging board-level hardware using schematics, datasheets, serial logs and laboratory instruments. Interested in developing reliable low-level firmware and embedded Linux products for intelligent transportation systems.

SKILLS SUMMARY

Programming and Firmware: C, C++ working knowledge, Python working knowledge, ESP-IDF, STM32 embedded development, MATLAB/Simulink, JSON

Embedded Platforms and Operating Systems: ESP32-S3, STM32, Raspberry Pi, embedded Linux, Linux command line, ARM-based systems

Interfaces and Networking: UART/RS232, I²C, CAN, I²S, SPI, Ethernet, Wi-Fi, MQTT, Modbus

Development Tools: Git/GitHub, Visual Studio Code, Altium Designer, Node-RED, CMake

Hardware Development and Testing: Board bring-up, schematic and datasheet interpretation, PCB design, sensor integration, oscilloscope, multimeter, power supply, soldering and serial debugging

Engineering AI tools: AI-assisted datasheet review, code troubleshooting, test-script development and technical documentation, with manual verification of generated results

WORK EXPERIENCE

Purus Power -- Conestoga SMART Research Centre Collaboration

Embedded Systems Engineering Co-op - Cambridge, ON

Sept 2025 - Present

- Developed and tested embedded edge devices for a multi-board wind-energy telemetry and control platform using ESP32 and Raspberry Pi hardware.
- Developed C firmware for ESP32-S3, integrating I²C, analog and digital sensors including the BME680, BNO085 and ADXL335, then transmitting telemetry through MQTT.
- Created MATLAB/Simulink models to simulate and validate sensor-driven control systems.
- Integrated embedded devices with Raspberry Pi/Linux services including Mosquitto MQTT and Node-RED, troubleshooting device-to-server comms. and real-time telemetry flow.
- Presented weekly progress updates to Purus Power and SMART Centre engineering teams.
- Performed hardware bring-up and interface debugging using schematics, component datasheets, serial logs, multimeters and oscilloscopes.
- Developed repeatable bench tests to validate sensor readings, communication interfaces and end-to-end telemetry behaviour before system integration.

Vortex Technology Group Collaboration with Conestoga SMART Research Centre

Research Assistant - Cambridge, ON

January 2025 -April 2025

- Reverse engineered an AC current-monitoring device to evaluate its sensing architecture, communication interfaces and suitability for battery test equipment.
- Interpreted component documentation and technical datasheets for current transformers and Modbus-to-Ethernet converters.
- Performed bench measurements and documented findings for engineering design and equipment-selection decisions.

EDUCATION

Electronic Systems Engineering (Co-op) - B. Eng

Conestoga College, Cambridge, ON
Sept. 2022 - Present



APPLIED PROJECTS

Smart Distributed Acoustic Calibration System, Conestoga College 2026

- Developed ESP-IDF firmware in C for ESP32-S3 acoustic sensor nodes, including I²S audio acquisition, sensor initialization, signal processing and network communication.
- Implemented 2048-point FFT processing, RMS and dBFS calculations, and periodic feature publication through MQTT over Wi-Fi.
- Integrated multiple embedded nodes with Raspberry Pi/Linux services including Mosquitto, Node-RED, SQLite and web-based visualization tools.
- Developed calibration and hardware-interface validation procedures using controlled acoustic inputs, serial diagnostics and laboratory measurement equipment.
- Diagnosed firmware, Wi-Fi and MQTT communication faults across the distributed multi-device system.
- Used Git/GitHub for branch-based firmware development, version control and integration of contributions across the project team.

Steam Punk Elevator Project VI, Conestoga College 2025

- Integrated and debugged CAN communication across STM32, Raspberry Pi, Arduino and motor-controller hardware in a multi-processor elevator control system.
- Developed C firmware for STM32-based control, safety interlocks and device communication.
- Coordinated dependencies between embedded control firmware, Linux-based Raspberry Pi services and a PHP/JSON user interface.
- Validated emergency-call and button-lockout functionality through system-level testing.

Nerf Battle Bot Mobile Robot Platform, Conestoga College 2024

- Designed, assembled and brought up custom Altium PCBs for motor-control, microcontroller and communication subsystems.
- Developed and debugged C firmware for DC, stepper and servo motor control, including hardware-interface and motion testing.
- Implemented and validated RS232/UART communication between the embedded controller and a Raspberry Pi 3 Linux system.